HEIN HTET

AI Engineer | Generative AI & LLM Systems

Bangkok, Thailand | +66 99 439 1816 | heinhtet.eie@gmail.com | linkedin.com/in/hein-htet-jeremy | github.com/heinhtet14

PROFESSIONAL SUMMARY

AI Engineer with 4+ years of hands-on experience across industry and research, building LLM products end to end: agent runtimes, real-time voice agents, LLM evaluation, AI infrastructure and inference on edge hardware. AI Architecture Lead at Magick Mind, leading the agent platform behind memory, persona, voice and robotics across 17 microservices. Research background in generative speech models (Gold Prize, Student Innovation Challenge Thailand 2024) and open-weight LLM benchmarking at SUTD. Strong systems foundation in Python, C/C++ and Rust.

TECHNICAL SKILLS

- **Generative AI & LLMs:** LLM agents, tool calling, multi-agent systems, agent memory, real-time voice agents (VAD, STT, LLM, TTS, barge-in), DSPy (signatures, ChainOfThought, ReAct), prompt engineering, LLM fine-tuning (Llama 2)
- **RAG & Knowledge:** Retrieval-augmented generation (RAG), graph RAG (LightRAG, RAG-Anything), document parsing (MinerU), vector search (pgvector), knowledge graphs (Neo4j), reranking
- **Evaluation:** LLM evaluation and benchmark design, LLM-as-judge, Recall@K, NDCG, ROUGE, BLEU, F1, embedding similarity
- **AI Infrastructure:** LiteLLM, OpenRouter, mixture-of-agents, Ray (KubeRay), Hatchet, per-tenant LLM spend control
- **Machine Learning:** PyTorch, TorchScript, ONNX Runtime, Ollama, Isaac Lab, reinforcement learning (PPO), WavLM, HiFi-GAN
- **Languages:** Python, C++, C, Rust, Go
- **Platform & Cloud:** Microservices, modular monoliths, Go (go-zero), Rust (Tokio, Axum, Tonic), FastAPI, REST APIs, gRPC, RabbitMQ, Centrifugo, etcd, Keycloak, Docker, Kubernetes, Helm, AWS S3, Google Cloud Run
- **Databases:** SQL, PostgreSQL, pgvector, MongoDB, Neo4j, Redis, SQLAlchemy, Alembic
- **DevOps & MLOps:** CI/CD (GitHub Actions), automated testing (pytest), DAST security scanning, OpenTelemetry, MLOps
- **Edge & Embedded:** NVIDIA Jetson Orin Nano, ESP32, STM32, ROS 2, CAN bus

WORK EXPERIENCE

Magick Mind | Bangkok, Thailand | Apr 2026 - Present

AI Architecture Lead & Team Co-Lead

AI agent infrastructure for developers: memory, persona and reasoning services with Python and Rust SDKs, working with any LLM provider, plus a companion app and ESP32 companion robots.

- **Architecture & leadership:** Lead AI architecture and co-lead a **~10-engineer team** across a **17-service platform** (Go, Python, Rust) spanning agents, personas, memory, knowledge base, inference and voice.
- **Microservices:** Designed the microservice boundaries and communication (gRPC, etcd service discovery, Centrifugo real-time messaging), then led consolidation into a gateway modular monolith and merged domain services, absorbing or retiring **7 services**.
- **AI infrastructure:** Designed the LLM gateway layer (LiteLLM virtual keys, OpenRouter) for provider-agnostic routing with per-tenant spend control, and a fast/smart inference path whose smart path runs several frontier models in parallel and aggregates their answers (mixture-of-agents); ran distributed memory ingestion on Ray (KubeRay) and Hatchet workers on Kubernetes.
- **Reasoning & evaluation:** Designed a deterministic recursive language-model architecture that replaces open-ended reasoning loops with a fixed combinator chain, and the protocol that benchmarks it against reference implementations on **4 public reasoning benchmarks**; also designed the agent-memory evaluation framework (LongMemEval, LLM-as-judge, retrieval and end-to-end metrics).
- **Real-time voice agents:** Led development of the Rust voice server: Gemini Live plus a cascaded VAD, STT, LLM and TTS pipeline with barge-in, language hints and noise gating, designed for up to **50 concurrent sessions** and **~500 ms time-to-first-audio**. LLM tool calls over the live stream drive **2 ESP32 companion robots'** motion and gestures, with safety stops.

- **Agent SDK:** Contributed Mindroid, the company's open-source Rust agent SDK (gated skill system, feature-gated builds for cloud or edge), and built its tool-calling engine (parallel tool calls, streaming tool rounds) and realtime voice layer (Gemini Live and OpenAI Realtime).
- **Devices & security:** Created the device backend (argon2 auth, rotating refresh tokens, robot pairing) and designed its token broker, which binds each device to its own persona-backed agent identity with scoped, short-lived credentials so tenant API keys never reach the hardware.
- **Embodied agents:** Built an LLM-to-robot adapter that exposes robot capabilities as agent tools over gRPC and Zenoh, validated on **3 MuJoCo** simulator (differential drive, Unitree Go2, Stretch) and inferenced on an SO-ARM101 arm.

Magick Robotics | Bangkok, Thailand | Mar 2025 - Mar 2026
Project Lead & Senior Robotics Engineer

- Led a small cross-disciplinary team building the Tet Hlan **12-DOF** bipedal humanoid; led RL training in Isaac Lab with PPO across **4096 parallel environments** (**44-dim** observation, 12 joint-position actions) and personally designed the reward terms.
- Took the policy from walking in simulation to the physical robot on NVIDIA Jetson Orin Nano at **50 Hz** through TorchScript and ONNX Runtime, reaching stable standing balance on hardware.
- Wrote the real-time C++ motor-control stack for **12 CubeMars motors** over **2 x 1 Mbit/s CAN** buses (**1 kHz thread** per bus, lock-free atomics, lookup-table ankle kinematics) and the **200 Hz** IMU node; integrated a Livox LiDAR with FAST-LIO and migrated the platform to ROS 2.

KMUTT Deep Learning Lab | Bangkok, Thailand | Aug 2022 - Jun 2025
Research Assistant

- Built an alaryngeal speech conversion system in PyTorch (kNN-VC, WavLM, HiFi-GAN) that **outperformed MaskCycleGAN-VC and SELM baselines** and won the **Gold Prize** at the Student Innovation Challenge Thailand 2024.
- Deployed it as a Flutter app with a FastAPI backend on Google Cloud Run, with end-to-end inference **under 3 seconds**.
- Trained and benchmarked sequence models over learned latent representations and designed the evaluation protocol; Teaching Assistant for Machine Learning and Deep Learning.

Singapore University of Technology and Design (SUTD) | Singapore | Jun 2024 - Aug 2024
Student Researcher

- Benchmarked **4 open-weight LLMs** (Llama 3, Phi-3, Mistral, Gemma) served locally with Ollama on qualitative coding tasks, using **3 custom evaluation metrics** (sentiment matching, embedding similarity, key-concept coverage) alongside ROUGE, BLEU and F1.

Panasonic Research Lab | Singapore | Oct 2023 - Jul 2024
Student Researcher

- Rewrote ORB-SLAM2 for constrained compute: built the numerical and optimization libraries (LU, Cholesky, SVD, Levenberg-Marquardt) from scratch and ported the **6-module** visual localization pipeline to dependency-free C.

Mu Space Corp | Bangkok, Thailand | Feb 2022 - Jul 2022
Embedded Engineering Intern

- Developed satellite firmware across **5 STM32 architectures**, including an ion thruster control subsystem.

EDUCATION

King Mongkut's University of Technology Thonburi (KMUTT) | Bangkok, Thailand | Jul 2025
B.Eng. Electronic and Infocommunication Engineering | GPA 3.71/4.00, First Class Honours

AWARDS & LEADERSHIP

- **Gold Prize**, Student Innovation Challenge Thailand 2024
- Founder, Clash of Robots, Myanmar's first national robotics competition (2018 - 2020)
- English: TOEIC 970/990